

JULIO LANDA

Mechatronics Engineer
Master's Degree in Renewable Energy



PROFESSIONAL EXPERIENCE

NC School Cuernavaca

Lead Physics and Mathematics Teacher · 2024-2025

Lead teacher for physics and mathematics courses.

IER, UNAM

Teaching Assistant · 2023-2025

Assisted with assignments, exams and project assessment for Introduction to Bioclimatic Design in the sixth semester of Renewable Energy Engineering, and served as teaching assistant for second-semester Linear Algebra.

IoT Developer Engineer · 2020 / 2024-present

Develop low-cost devices using open-source hardware and software, integrated with IoT platforms for the instrumentation project of the new Renewable Energy Engineering building at IER, UNAM.

CNEER

Workshop Coordinator · 2023

Planned, organized and coordinated workshops for the 10th edition of CNEER, the National Congress of Renewable Energy Students.

CNEER

General co-coordinator · 2026

General Coordinator of the National Congress of Renewable Energy Students (CNEER) 2026.

DMC Ingenieria

Electrical Engineer · 2022

Provided maintenance to electric motors and control panels as a contractor for Cementos Moctezuma.

Bombas y Servicios Corbala

Electrical Engineer · 2021

Maintained and installed deep-well water pumps, and designed and installed control systems and electrical panels.

Alucaps

Maintenance Technician · 2020

Applied maintenance plans to company machinery, updated and programmed production-line control systems using PLCs, and assembled electrical panels.

EDUCATION

Master's in Engineering - Energy Area

Instituto de Energias Renovables, UNAM · 2022-2025
Morelos, Mexico.

Bachelor of Mechatronics Engineering

Universidad Tecnologica de Emiliano Zapata · 2018-2020
Emiliano Zapata, Morelos, Mexico.

Higher University Technician in Mechatronics, Automation Area

Universidad Tecnologica de Emiliano Zapata · 2016-2018
Emiliano Zapata, Morelos, Mexico.

CONTACT

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 JulioLanda4

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LANGUAGES

- Spanish: native
- English: upper-intermediate/advanced

SKILLS

- **Instrumentation:** instrument design and calibration.
- **Maintenance:** preventive and corrective maintenance.
- **Electrical panels:** design and assembly of electrical systems.
- **Programming:** Python, C/C++, Arduino and LabVIEW.
- **Automation:** design and development of industrial control systems.
- **Mechanical design:** CAD modeling with SolidWorks and FreeCAD.
- **Data analysis:** Python, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Seaborn and Plotly.
- **Artificial intelligence:** machine learning, neural networks and genetic algorithms.
- **IoT:** development with ESP32/Arduino and platform integration.
- **Electronics:** circuit design and automation projects.
- **PLCs:** control system programming.

PUBLICATIONS

IoT smartwatch based on open technologies for the collection of thermal comfort data

HardwareX · 2025

DOI: <https://doi.org/10.1016/j.ohx.2025.e00633>

IoT smartwatch based on open technologies for the collection of thermal comfort data

Master's thesis · 2025

Available at: https://altamarmx.github.io/reloj_inteligente/

Project licensed under GPL v3.0: https://github.com/JulioLanda4/IoT_Relaj_Confort/tree/v1.0.0

CERTIFICATIONS, COURSES, HONORS & AWARDS

Certifications

- **Certified SolidWorks Professional (CSWP)** · 2019.
- **Google IT Automation with Python** · 2022.
- **LabVIEW Associate Developer** · 2019.

Courses

- **Macrotraining in Artificial Intelligence (MeIA)** · 2023.
- **Python for Engineering (UNAM, Coursera)** · 2023.

Honors & Awards

- **Recognition as speaker** · 2024. Sustainable Buildings Symposium: Simulations and Digitalization for Decarbonization.
- **Honorable Mention** · 2025. XXXV CUAM-ACMor Research Congress, Physics-Mathematics category, project "Green Motion: How Do You Move?".
- **Recognition** · 2026. Participation as part of the organizing committee for the Hackathon HackODS 2026, organized by the Institute of Renewable Energy and the Institute of Engineering at UNAM, from January 12 to April 24, 2026.
- **1st place** · 2019. 3rd Mathematics Olympiad, Universidad Tecnológica de Emiliano Zapata.
- **Honorable Mention** · 2023. XXXIII CUAM-ACMor Research Congress, Physics-Mathematics category, project "Ecological Greenhouse from Plastic Waste".

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Nota: La inserción se realiza justo después del encabezado de la sección de reconocimientos en ambos archivos, manteniendo el orden cronológico inverso.

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- **Honorable Mention** · 2023. XXXIII CUAM-ACMor Research Congress, Physics-Mathematics category, project "Ecological Greenhouse from Plastic Waste".
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